

Ian Tai Ahn

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PROFESSIONAL SUMMARY

Software Engineer with 2.5+ years supporting large-scale defense programs at Hill AFB (Lockheed Martin, Pratt & Whitney). Expertise in on-premises Kubernetes clusters, CI/CD pipelines, full stack development and microservices architecture. M.S. in Data Science with a strong foundation in machine learning and statistical modeling. Active Secret clearance.

EXPERIENCE

Software Engineer – DevOps & Platform Engineering

January 2025 – Present

Lockheed Martin | Hill Air Force Base, UT

- Designed and operated a globally deployed Kubernetes service mesh using Istio, supporting 30+ distributed microservices across multiple enclaves
- Automated policy enforcement across all services using Helm, Kyverno, and Ansible, reducing manual deployment errors by an estimated 40%
- Integrated testing and CI workflows to ensure >80% system test coverage, catching defects before production deployments
- Led architecture decisions for a microservices-based application serving as a core component of the program's mission system
- Owned containerization strategy and authored Helm charts to standardize deployment and configuration management across the entire service portfolio in an agile environment
- Served as Scrum Master and Technical Lead, coordinating sprint planning, backlog grooming, and cross-team dependencies

Software Engineer – Systems Testing & Automation

August 2023 – December 2024

Pratt & Whitney | Hill Air Force Base, UT

- Developed and maintained automated system tests using Robot Framework, validating complex integrated systems and reducing manual test execution time
- Improved defect detection and system reliability through structured test design, contributing to comprehensive regression coverage across releases
- Authored a CSV parser that automated a manual 20–30 minute task, reducing completion time to seconds and saving an estimated 10 hours per month

TECHNICAL SKILLS

Languages & Frameworks: Python, R, JavaScript, TypeScript, SQL, PostgreSQL, Java, React, Django, Tailwind, Robot Framework

ML & Statistics: PyTorch, TensorFlow, scikit-learn, Two-Stage Least Squares, Instrumental Variables, OLS, statistical modeling

Data: pandas, NumPy, data cleaning, feature engineering, dataset preparation, re-labeling

LLMs & RAG: Embeddings, vector databases (FAISS), end-to-end Retrieval-Augmented Generation (RAG) pipelines

DevOps & Cloud: Kubernetes (K8s), Docker, Helm, Istio, Ansible, Kyverno, Amazon Web Services (AWS), Linux

Other: Git, REST APIs, Agile

PROJECTS

RAG Pipeline | Python, React, FAISS | <https://github.com/ianTaiAhn/rag/>

- Built an end-to-end RAG pipeline covering document ingestion, preprocessing, chunking, embedding generation, vector storage, and semantic retrieval
- Designed a multi-index FAISS system with full CRUD support and explicit index selection for isolated, context-aware retrieval

EDUCATION

M.S., Data Science | Graduate Certificate, Computational Data Science & Machine Learning 2025

Weber State University, Ogden, UT, GPA 3.85

B.S., Computer Science 2023

Weber State University, Ogden, UT, GPA 3.93